AI Meal Planner

One meal. Clear nutrition. A shopping list.

MULTI-AGENT MEAL PLANNING

Say what you crave. Get a meal that fits you.

A FastAPI service sizes one meal to your body and goal, looks up each ingredient, labels any estimates, balances the macros and prices the shopping list. Each number says how it was produced.

You → Craving → Your plan

For your craving “chicken fried rice”

Chicken Fried Rice

Close menu match · On target · Some nutrition estimated

This meal	Protein	Carbs	Fat
629 kcal Target 625 kcal	42 g	71 g	18 g

Protein	27% · range 10–35%
Carbs	46% · range 45–65%
Fat	27% · range 20–35%

HOW A PLAN IS MADE

Four agents, one typed response

1

Calorie expenditure

A gradient-boosted model trained on 750,000 Kaggle rows predicts exercise calories, which feed a daily budget and the meal's target.

Exercise model R^2 0.996

2

Meal recommendation

Allergy, diet and health filters run first, then local vector RAG retrieves from 34 curated recipes and rescales portions.

Works with no LLM key

3

Nutrition verification

Each ingredient is checked against USDA FoodData Central, with FatSecret and local tables as cached fallbacks.

Estimates are labelled

4

Supermarket

Ingredients map to grocery products with an estimated cost and a tick-off list, marked as estimates.

Shopping list included

MEAL QUALITY, MEASURED

From 10 to 72 of 72 plans on target

72/72

plans in a fixed, deterministic benchmark land within 10% of their calorie target with protein, carbs and fat all inside the adult AMDR ranges.

Baseline	10
Per-ingredient balancing	41
Honest fallback gate	47
Cooking fat + rounding retry	72

0.3%
median calorie deviation

5 / 5
guardrail cases hold, including refusals

72
requests: 4 body types, 4 diet and health limits

BUILT TO BE TRUSTED

Production habits on a study project

Typed API contract
FastAPI + Pydantic; OpenAPI committed and checked in CI

Tested
590+ Python tests at 95% coverage, plus React tests

Secured
API keys, rate limits and a hosted mode with no history

Deployable
Containers and a Cloud Run workflow with keyless auth

Observable
OpenTelemetry tracing that keeps health data out of traces

Two clients
A live Streamlit demo and a React + Tailwind app

Live demo · tuannm3812-ai-meal-planner.streamlit.app
Code · github.com/tuannm3812/ai-meal-planner

Python · FastAPI · scikit-learn · RAG · USDA API · React · Streamlit
A study project. Not medical advice.